

KARTHIK CHAKALA

Software Engineer | Backend & System Design

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PROFESSIONAL SUMMARY

CS undergraduate at IIIT Kurnool with strong foundations in Data Structures & Algorithms, Object-Oriented Programming, Database Systems, and Software Engineering. Hands-on experience designing and developing scalable backend and full-stack applications using C++, Java, Node.js, React.js, and PostgreSQL. Familiar with High-Level and Low-Level System Design, SOLID principles, design patterns, REST API design, and scalable application architecture. Solved 300+ DSA problems with a LeetCode rating of 1600+. Strong problem-solving, debugging, and software development skills with a focus on writing reliable, maintainable, and high-quality code.

TECHNICAL SKILLS

- **Languages:** C++, Java, Python, JavaScript
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks
- **System Design:** HLD, LLD, SOLID Principles, Design Patterns, UML, API Design, Scalability, Caching, Database Design
- **Backend:** Node.js, Express.js, REST APIs, JWT Authentication, Socket.IO, Node-Cron
- **Databases:** PostgreSQL, MongoDB, MySQL, Oracle DB, SQLite, Supabase
- **Engineering:** Linux, Docker, Git, GitHub, Postman, Logging, Debugging, Testing
- **Frontend:** React.js, Next.js, Redux, Tailwind CSS
- **DSA Practice:** LeetCode 1600+ | 300+ problems solved across LeetCode and GeeksforGeeks

PROJECTS

Farm to Home | Agri-Commerce Platform (Dec 2025 – Mar 2026)

- Architected a hyperlocal agri-commerce platform using React.js, Node.js, Express.js, and Supabase, connecting farmers to consumers within a 5–7 km radius via distance-formula-based filtering.
- Designed 60+ RESTful APIs across cart, orders, inventory, and expert consultation workflows with rate limiting, input validation, and Winston/Morgan logging.
- Enabled live farmer-consumer chat, expert sessions, and order tracking using Socket.IO, webRTC; automated subscription billing with Node-Cron and payments via Razorpay.
- Delivered an AI agriculture suite with pest detection, a crop growth simulator, and profitability analysis across a microservice architecture.

Hospital Management System | Project Lead (Jul 2025 – Oct 2025)

- Spearheaded a 4-member Agile team to plan, develop, and ship a Hospital Management System across the full development lifecycle.
- Built the backend with Node.js, Express.js, and Supabase; developed the frontend using Next.js and React.js.
- Delivered 30+ permission-based REST APIs spanning appointments, EMR, billing, lab, pharmacy, inventory, and home-visit modules.
- Secured 20+ endpoints with token-based authentication and integrated Cloudinary, Multer, Nodemailer, and Razorpay.

E-Learning Platform | Full-Stack Developer (Aug 2025 – Nov 2025)

- Shipped an e-learning platform with 5+ modules covering authentication, payments, lessons, quizzes, and live competitions.
- Developed a scalable Node.js and Express.js backend with 40+ endpoints and access control for 3 distinct user roles.
- Powered live competitive quizzes using Socket.IO namespaces with transactional course purchase workflows.
- Enforced data integrity via HMAC webhook verification, parameterized SQL queries, and 10+ database constraints and triggers.

EDUCATION

- **Indian Institute of Information Technology, Kurnool** — B.Tech, Computer Science and Engineering (2023–2027) | CGPA: 8.4/10
- **Amaravathi Jr College** — MPC | JEE Percentile: 97.4%

RELEVANT COURSEWORK

- Data Structures and Algorithms • Operating Systems • Database Management Systems • Computer Networks • Object-Oriented Programming

CERTIFICATIONS & COMPETITIVE PROGRAMMING

- **Certifications:** IITB — Python, C, Linux | NPTEL — Cloud Computing
- **Competitive Programming:** LeetCode 1600+ | CodeChef 3★ | 20+ rated contests